

VIDEO SCRIPTS

UNIT 1

▶ China plans revival of Silk Road trade routes

Commentator: We're up in the mountains near the border with Kazakhstan on China's far western frontier. The area we're travelling through is pretty remote, close to the furthest point from an ocean on the planet. This is the land of the old Silk Road where caravans of merchants trekked more than a thousand years ago at the height of China's imperial past. It's now at the heart of the country's plan to rebuild those ancient trading routes over land and by sea in one of the most expensive development projects ever attempted. This is a good place to come to understand the scale of those ambitions. Five years ago, there was almost nothing here. All of this has been built from scratch, including this brand new dry port intended as a key logistics hub for road and rail. Alongside a vast new free trade zone. Mr Guo has big dreams for this project. He tells me Korgas could be the new Shenzhen or Shanghai.

Guo Jianbin: I think in three or four years, you will see the sky in Korgas will be more blue, and the city will become more beautiful and international trade will play a more key role.

Commentator: This is the 'build it and they will come' approach to infrastructure and trade, build the roads and rail links out into the regions' developing countries and the trade will follow, so the theory goes.

This huge cargo train is lumbering past us now on its way from China to Europe and this is very much the image of the modern Silk Road the Chinese government wants to project. But there's more going on here than just investment in infrastructure. This project is as much about politics as it is about economics.

Ethan Cramer-Flood studies the project for an economic think tank and sees it as being driven primarily by China's domestic economic objectives and its broader geo-political goals.

Ethan Cramer-Flood: This is absolutely first and foremost err ... a soft power initiative where it's clearly err ... China has the leaders ... leadership

in Beijing and Beijing envisions the One Belt One Road project as being a significant, maybe the number one most significant part of Xi Jinping's err ... so called 'China dream' and vision of ... of supporting the emergence of China onto the global stage as the next great superpower.

UNIT 2

▶ Disadvantaged children take part in trial of private tutor app

Commentator: GCSE Maths, it's tough enough even when you're in the classroom and the teacher's there to answer your questions. But what about when you're doing your homework and there's no one around who can help? These pupils in Tower Hamlets are trying out a new app. It's giving free instant Maths help to those who might not otherwise be able to afford private tuition.

Peter Kirby: We know that over 90% of teenagers' digital communication is now text-based, so we wanted to design something that would meet them where they were.

Sheikh Fahmida Musarrat: If you, or your Mum or Dad don't know, it's like the next alternative. You don't need to go and look for tutors, and you don't have to like, stay behind in class just to catch up on work.

Commentator: Catching up outside the classroom is extra important these days; in London almost half of secondary school pupils have now received private tuition.

Dr Lee Elliot-Major: The gap between the 'haves' and the 'have-nots' if you like in education is widening, outside the school gate, not just in schools. So we're really concerned about that, and that's why we think that these sorts of agencies that provide tutoring should look at ways at helping those children from poor backgrounds.

Commentator: Oaklands School here in Tower Hamlets is one in five across the capital now piloting this new app, the others are in Lewisham, Brent, Lambeth and Westminster and the idea is, if it's successful, to roll the app out across the whole of the UK.

Teachers here say they'd welcome anything that helps their pupils get ahead.

Simon Ramsay: Not all of our students can afford private tutors ... erm, and this is a way that circumvents that and allows them to access err ... qualified err ... tutors in their own time so in terms of levelling the play ... playing field err ... yeah it's a great opportunity.

UNIT 3

 **Robot revolution: robotic surgery is on the rise**

Commentator: They don't get tired, their hands don't tremble and their patients make a faster recovery. Robot surgeons are increasingly used in NHS hospitals, but the revolution has only just begun.

Gauthier Gras: It is a sort of see-through vision.

Commentator: Robotic technology will in future give surgeons super-human powers. This probe being developed at Imperial College London not only warns a brain surgeon when they're pressing too hard on delicate tissue, but it also reveals the structures that lie beneath, using scans taken before the operation.

Gauthier Gras: If you're closing in on this aneurysm you'd really want to be aware of the forces that are applied. Err ... it's, it's very easy to ... to apply too much force, and then suddenly rupture it. You're able to warn the surgeon ahead of time to prevent damage.

Commentator: Shrinking technology is allowing scientists to make robotic tools too small to see with the naked eye. These are less than half the width of a human hair, and dexterous enough to move a microscopic ball. It could one day be possible to operate on individual cells at the earliest stage of disease.

Professor Guang-Zhong Yang: The type of robot we are making actually getting smaller and smaller can be actually delivered in the needle form. So therefore, you don't need to make any incisions. So, they can actually follow the err ... curved anatomical pathways, reaching to the side.

Dan Melville: If I think about flicking up lightly, it opens like that.

Commentator: Robotics of a different kind, and for Dan Melville it's life changing.

Dan Melville: So, this is me, err, trying to test it out and see how it goes. This is the first time actually testing it, so, err ... good luck to myself.

Commentator: By tensing his arm muscles he can control his bionic digits, changing his grip to pick up different objects. It's sophisticated technology, but it's being printed in just a few hours, sized to fit and in his choice of colour.

Dan Melville: You can customize it, you can change the colour so if, err ... a girl wanted black and orange and red, you can have whatever you want and you would just and I ... I personally think it's a fantastic opportunity to just kind of show who you really are on your arm. So, it makes it more personalized for you as well.

Commentator: High-end bionic hands can cost up to £90,000, beyond the reach of the NHS, but Open Bionics hopes to launch its own version by the end of next year, the price of less than £2,000.

Sammy Payne: All of the grip patterns that the advanced bionic hands can do, all of the movements we are replicating and the manufacturing techniques that they're ... they're using, um ... is a very expensive manufacturing process. Whereas we have desktop 3D printers that just print using plastic filaments. I mean you couldn't ... they're worlds apart, the manufacturing methods. Um so that's how we can be so low cost.

Commentator: Patients and surgeons enhanced by robotic technology. We're entering the age of the cyborg.

UNIT 4

 **Population and water**

Narrator: We call our Earth 'the blue planet' because about 70% of the Earth's surface is covered in water. But most of that is in the oceans and seas. Just 2.5% is fresh water, and only 1% of that is available for human use. The rest is locked up in mountain passes and the Earth's polar ice caps. But there's another fact we need to understand about water.

Brian Richter: Well, there's no more water on the planet than there was when life first appeared on Earth. It changes its distribution, there's more water in different parts of the world than there were hundreds or thousands of years ago, but it's still exactly the same amount of water that's been here always.

Narrator: We use over half of all the available fresh water in the world to serve our needs: to transform deserts into fields, to produce energy from rivers and to build cities in some of the driest regions on the planet. But despite our creativity, there are many who have difficulty getting enough of this basic resource.

Brian Richter: More than a billion people on the planet already lack access to safe, clean drinking water. And we know things are going to get more difficult as the population continues to grow. Within the next 20 years, as much as half of the world's population will live in areas of water stress.

Narrator: Many water shortages are the result of poor infrastructure, politics, poverty or simply living in a dry part of the world. But more and more, they are due to increasing populations. Mexico City, for example, benefits from heavy annual rainfall. But its water system is stressed from supplying water to its 20 million inhabitants. The issue is the combination of leaks in the system and the fact that backup reservoirs are running dry. In Mexico City, shops that sell water for people's daily needs are becoming more and more common. But the water we use at home is only a small percentage of the total amount of water we consume. That's because of the huge amounts used by farms and factories.

Brian Richter: We may know where the water out of our tap comes from, but we very seldom know where the water that went into our can of cola or into the shirt that we're wearing on our back, where those goods were produced and how much water it required, and what the consequences were for the natural systems in those areas and for the local communities that are dependent upon that same water.

So for example, the cup of coffee that you may have in the morning requires on the order of 120 litres just to produce the coffee and bring it to your table. A hamburger, 8,000 litres of water, to produce enough water to grow the cotton in my shirt is 3,000 litres as well.

Narrator: The influence of humans on the world's fresh water systems is so significant that it can be seen from space. The Aral Sea, the fresh water lake in central Asia, once

covered more than 25,000 square miles. But in the last 40 years, it has lost 90% of its water, with most of it going to support cotton farms. Lake Chad, on the southern side of the Sahara Desert, is now one tenth of its normal size due to drought and overuse. Yet, 30 million people still depend on it.

UNIT 5

Government grants for warmer, cheaper housing

Commentator: Britain's got some of the chilliest buildings in Europe, £1 in every four we spend on heating is wasted due to things like poor insulation. Today the government announced it will extend grants for insulation work like this for warmer, cheaper housing for people who can least afford it.

Gareth Redmond-King: Emissions from homes have started to go up now, and that's a huge problem given the ... the scale of the challenge to reduce our emissions by 80% by 2050. Um ... so at the current rate of err ... improving the energy efficiency of our homes we've calculated it will take us over a hundred years err ... to cut emissions from our homes.

Commentator: So, this is the sort of thing campaigners welcome, but worry whether the government will deliver on, with good reason. Cygnus Homes in Cambridgeshire build to the 'passive house' standard, engineered and insulated in such a way that just by living in the house it achieves a comfortable temperature, no extra heating required.

The energy bill for this house is about 75% less than the national average, and it can be built in less time by fewer workers than the traditional home. It shows that the construction industry can deliver the low carbon houses that Britain desperately needs, in very little time. So, why aren't we building them? Well because a couple of years ago in response to intense lobbying the government dropped something called the 'zero carbon home standard'.

As a result, housing developments with very efficient new homes like this one in Essex are the rarity rather than the norm. Even though innovative architects and developers say they can deliver them at a similar cost to more wasteful traditional homes.

So, the decision to scrap the zero carbon homes target, did that set everyone back do you think?

Tom Dollard: Undoubtedly yeah, it, um ... the industry had a clear target which was meant to be zero carbon homes at 2016 ... that was scrapped and ... and then there was a lot of confusion, there still is a lot of confusion, so ...

Reporter: So now there's an opportunity to put that back though, right?

Tom Dollard: Yeah, we need a clear target, um ... simple policy to set house builders and developers a standard for zero carbon homes or low carbon homes whatever it needs ... it just needs to be a standard.

Reporter: And you're saying in some ways it shows that it can be done, it's within our grasp, it's not impossible.

Tom Dollard: Exactly.

Commentator: Today's strategy includes most of the things needed to deliver a truly greener economy. But, say critics, this government mustn't repeat the mistakes of others by giving up on green measures that long-term benefit consumers and the environment.

UNIT 6

The power of the wind

Narrator: Wind power provides only about five percent of the world's energy needs today, but many environmental experts believe that figure could rise to almost 20% by 2030. Wind generates almost half a million megawatts of power—that's a 19% jump in just 5 years. The UK is in a particularly good position to develop wind power because it is so, well, windy. At present, it ranks sixth in terms of capacity but is investing heavily in this renewable energy source. The percentage of electricity in the UK generated from wind power overtook that of coal, once Britain's primary source of energy, for the first time in 2016. The result was the lowest carbon emission rate since the 1920s.

The blades of the wind turbine turn a shaft, which passes the energy to a generator, which in turn transforms this energy into electricity. The wind is free, but the cost of power generation

and storage actually makes wind power more expensive than traditional fossil fuel sources like coal, oil and gas. Smart technology is steadily bringing these costs down, however, and this trend is expected to continue.

There are two different types of wind power – onshore and offshore. Onshore wind farms are cheaper to build and maintain, but many people don't want windfarms in their areas. The turbines are quite noisy, and many people say they spoil the landscape.

Offshore windfarms, in contrast, are located in the sea, where few people care about their noise or appearance. The UK is a leader in offshore wind power. The London Array, off the east coast of England in the North Sea, is the largest offshore wind farm in the world.

The UK is also home to the world's first floating windfarm, off the northeast coast of Scotland.

Although there is considerable public support for wind power, there is also opposition to these large, noisy devices. It's also important to remember that all of them are subject to changes in weather patterns. There are windy days and not-so-windy days. There are also not-so-windy years. Denmark, a leader in wind power, saw its wind power level drop in 2016 because of a lack of wind. Nevertheless, the shift to renewable energy sources like wind is likely to continue unabated.

Energy analyst: If you look at a more flexible, more dynamic energy system, we're saying more electric vehicles on ... going out into the 2030s, then the backbone of that energy system should be cheap renewables. That is going to be the way to keep bills ... bills low and you won't find an energy analyst out there that disagrees with that vision of the future.

UNIT 7

Beijing Art Zone

Narrator: This was once a vast state-run military factory on the outskirts of Beijing, employing as many as 20,000 workers, complete with housing for the workers, athletic facilities, a hospital, and even its own orchestra. But that was the 1950s. Today, this factory complex presents a more peaceful and dramatically different face.

798 Art Zone became a space where artists could live and create art in a supportive community that included painters, sculptors, fashion designers, photographers and film directors. It attracted well known Chinese artists, such as Sui Jianguo, whose dinosaurs are popular with visitors.

Visitors stop to examine and photograph these Red Metal Men, not quite sure if they are singing, shouting, staring in surprise or just hoping to catch some raindrops!

Although the Art Zone began as a place for outsiders, artists whose work was not seen in traditional museums, it has become a part of the mainstream arts scene, a popular attraction with both local visitors and international tourists. The centre hosts art exhibitions, annual film festivals, fashion shows, art auctions and theatre productions, and it has all the services common to other tourist destinations, including gift shops and cafés like this one, located in an old train station that used to transport materials to and from the original factory. It's free to the public and tens of thousands of visitors enjoy the centre every day.

When it began, the 798 Art Zone provided inexpensive living and work space for struggling artists. Today the rents are high and only the most successful artists can afford to show their work here. The struggling artists have moved on to new locations – perhaps to the next 798.

Commentator: This group goes out for an hour at a time with two different routes for quicker and slower walkers, and it's precisely what health experts say all adults need to do, regular gentle exercise, even just ten minutes a day can help prevent cancer, heart disease and poor mental health.

Dr Jenny Harries: No fancy kit, no expensive gym memberships, sensible pair of shoes and off you go um ... and you can just err ... do it when you're taking the dog for a walk, do it if you're taking the kids to school, err ... do it if you're walking to work, so really easy but it really does have huge health benefits.

Commentator: Research by Public Health England suggests that we're taking less and less exercise, but people in the UK are 20% less active than in the 1960s. They say just a few minutes brisk walking every day could reverse that trend and would come with other benefits.

Barry Hazel: There are three benefits a) the exercise, b) the fresh air and c) the companionship and I think that's also very important because a lot of these people are widows or widowers and they might not see many people during the week.

Commentator: And these walkers say the other advantage of exercise is it makes the cup of tea afterwards taste even better.

UNIT 8

The retired men and women who love to walk

Commentator: Seventeen years ago, a group of newly retired men and women in Salford decided to meet up every week and go for a walk, once a week for seventeen years and today they're still going. The handful of original walkers has grown to fifty or so and they're loving it.

Wilma Harriman: When I go on a walk, I feel energized and I feel as if I can cope with things.

Mary Walker: My husband tried his best to get me swimming which I believe is the best form of exercise but um ... I'm frightened of the water.